Toward a four-dimensional Yang–Mills construction

Notebook 07: locality and refinement limit of the flat Gaussian

Research working document

16 October 2026

Abstract

We derive uniform exponential spatial decay for the flat quadratic effective action and its constrained Gaussian covariance. The action is a gauge quotient of an averaged Feynman covariance. An exact algebraic cancellation removes its apparent complex massless poles. A discrete Parseval estimate controls the growing number of fine Fourier aliases and supplies a complex momentum strip independent of refinement. The fixed local gauge penalty and pivot insertion then combine this locality with the spectral gap proved in Notebook 06. We also prove convergence to an explicit continuum sinc-alias symbol at rate $O(M^{-2})$ in an exponentially weighted kernel norm, uniformly over compatible periodic volumes. The refinement and thermodynamic limits commute for this free reference. These results supply the flat baseline required by the earlier Gaussian cutoff theorems. They do not control the actual complex-background perturbations or prove a nonlinear continuum Yang–Mills limit.

Status. Research proof checked by separate internal derivations and audits. No external certification or novelty claim is made. The flat variational model is defined explicitly below. Notebook 15 now supplies the smooth ordinary-constraint minimizer required by Notebook 06, and Notebook 17 closes its homogeneous flat Hessian identification. Section 8 records the precise deposited coordinate match.

1 The result and its scope

Fix dimension d and odd next-step blocking factor $L \geq 3$. Let $M = L^k$ be the refinement from physical spacing $\eta = M^{-1}$ to spacing one. More generally, any odd positive M is allowed. Use the straight-string average and physical volume norms of Notebook 06, and define

$$q_M(B) = \inf_{Q_M^{\mathrm{st}} A = B} \|d_\eta A\|_\eta^2 .$$

For finite periodic lattices this is a finite-dimensional quadratic form. Its translation-invariant kernel and infinite-volume version will follow from the Fourier calculation below.

Let T_L be the averaged rooted-path map within fixed L-blocks, $S_L = Q_L^\flat$ the next-step rooted constraint, and K_L the actual flat insertion of pivot bonds into the retained nonpivot coordinates. The reduced precision is

$$A_{M,0} = K_L^*[q_M + T_L^* T_L]K_L. \tag{1}$$

All stars here are real Hilbert-space adjoints in the fixed volume conventions. Their holomorphic continuations are bilinear transposes, not variable-dependent complex conjugations.

Theorem 1.1. *There are positive constants ρ, C_q, depending only on d, such that the unit-lattice action kernel satisfies*

$$\|q_M(x)\| \leq C_q e^{-\rho|x|_1}$$

uniformly in M. There are positive c, C, depending only on fixed d, L and the fixed color norm, such that the retained-coordinate covariance satisfies

$$\left\| A_{M,0}^{-1}(n,\alpha;m,\beta) \right\| \le Ce^{-c|n-m|_1}.$$

Here n, m are L-block indices and α, β are retained bond labels within a block. Finite periodic kernels are exact periodizations, with uniform exponential bounds in periodic distance.

The theorem concerns the homogeneous full-block quadratic problem. A boundary upper bound, such as that of Notebook 06, does not imply that a boundary effective covariance is a restriction of this one. An arbitrary retained-history or nonflat operator needs its own match.

2 The exact alias formula and the gauge quotient

Let $I_M = \{-(M-1)/2, \dots, (M-1)/2\}$ and, for coarse momentum z, write $p_r = z + 2\pi r$, $r \in I_M^d$. Define the entire multipliers

$$a_M(p) = \prod_{\nu=1}^{d} \frac{1}{M} \sum_{s \in I_M} e^{ip_\nu s/M}, \qquad \ell_{M,\mu}(p) = \frac{1}{M} \sum_{s=0}^{M-1} e^{ip_\mu s/M},$$

$$\omega_\mu(p) = a_M(p)\ell_{M,\mu}(p).$$

This is the actual one-form averaging multiplier. Its longitudinal factor occurs in addition to the centered block factor in that direction. A scalar block average alone would be the wrong operator.

Write

$$u_{r,\mu}^{\pm}(z) = M(e^{\pm ip_{r,\mu}/M} - 1), \qquad P_r(z) = \sum_\mu u_{r,\mu}^{+}(z)u_{r,\mu}^{-}(z) = 4M^2 \sum_\mu \sin^2(p_{r,\mu}/(2M)),$$

$$d_\mu^{\pm}(z) = e^{\pm iz_\mu} - 1, \qquad b_\mu(z) = d_\mu^{+}(z)d_\mu^{-}(z) = 4\sin^2(z_\mu/2).$$

Minus means analytic substitution of the negative argument. It agrees with conjugation on real momenta only. String telescoping gives

$$\omega_\mu(p_r)u_{r,\mu}^{+}(z) = a_M(p_r)d_\mu^{+}(z), \tag{2}$$

and the corresponding minus identity.

With physical-volume Parseval normalization, $B_\mu(z) = \sum_r \omega_\mu(p_r)A_{r,\mu}$. An embedded plane wave has the same fine-volume and coarse-volume norm, so no extra M^d factor enters the alias covariance.

For real $z \neq 0$ in the central Brillouin zone, all $P_r > 0$. The fine Maxwell energy in one alias is

$$P_r|A_r|^2 - |(u_r^{+})^* A_r|^2 = \inf_\lambda P_r|A_r + u_r^{+}\lambda|^2.$$

By (2), fine gradients produce precisely coarse gradients. They produce every scalar multiple of d^{+} because the central $a_M(z)$ is nonzero on this real zone. Thus Maxwell minimization equals Feynman minimization with the coarse representative allowed to vary by a gradient. The averaged Feynman covariance is diagonal:

$$W(z) = \mathrm{diag}(c_\mu(z)), \qquad c_\mu(z) = \sum_{r \in I_M^d} \frac{\omega_\mu(p_r)\omega_\mu(-p_r)}{P_r(z)}. \tag{3}$$

For real momenta its numerators are squared absolute values. Minimizing the positive Feynman form at prescribed coarse datum B' gives $B'^{*}W^{-1}B'$. A subsequent minimization over $B' = B + d^{+}\lambda$ gives the full-space precision

$$q_M = W^{-1} - W^{-1}d^{+}(d^{-}W^{-1}d^{+})^{-1}d^{-}W^{-1}. \tag{4}$$

This also follows by the ordinary scalar Schur complement. Neither W^{-1} nor the transverse compression of W^{-1} alone is generally this quotient.

At $z = 0$, every nonzero alias has $a_M(2\pi r) = 0$. The average reads fine constants, which realize every constant coarse field with zero Maxwell energy. Therefore

$$q_M(0) = 0. \tag{5}$$

The harmonic sector is handled explicitly; no inverse of its zero Feynman eigenvalue is assigned.

3 Cancellation of the complex massless denominator

Separate the central alias and define

$$A(z) = a_M(z)a_M(-z), \quad g_\mu(z) = \omega_\mu(z)\omega_\mu(-z), \quad R_\mu(z) = \sum_{r\neq 0} \frac{\omega_\mu(p_r)\omega_\mu(-p_r)}{P_r(z)},$$

$$D_\mu = g_\mu + P_0 R_\mu, \qquad h_\mu = D_\mu^{-1}.$$

The central gradient identity implies

$$\sum_\mu \frac{b_\mu}{g_\mu} = \frac{P_0}{A}. \tag{6}$$

All removable one-variable factors are understood analytically. Set

$$E = A^{-1} - \sum_\mu \frac{b_\mu R_\mu}{g_\mu D_\mu}. \tag{7}$$

Subtracting $D_\mu^{-1} - g_\mu^{-1}$ in (6) proves exactly

$$\sum_\mu b_\mu h_\mu = P_0 E.$$

On the original real nonzero set, $W^{-1} = P_0\,\mathrm{diag}(h_\mu)$. Substitution in (4) and cancellation gives

$$(q_M)_{\mu\nu}(z) = P_0(z)h_\mu(z)\delta_{\mu\nu} - \frac{h_\mu(z)d_\mu^{+}(z)d_\nu^{-}(z)h_\nu(z)}{E(z)}. \tag{8}$$

There is no inverse power of P_0 left. This removes its entire complex null set, rather than just the real zero at the origin. We will prove that the remaining denominators are uniformly nonzero on a complex strip. The resulting formula is zero at $z = 0$, agreeing with (5). For $M = 1$, $A = g_\mu = D_\mu = E = 1$ and $R_\mu = 0$, giving the usual finite-range Maxwell precision.

4 A complex momentum strip independent of refinement

4.1 Alias Parseval prevents a factor M^d

Discrete orthogonality gives, for real x, y in one coordinate,

$$\sum_{r\in I_M}\left|\frac{1}{M}\sum_{s\in I_M} e^{i(x+2\pi r+iy)s/M}\right|^2 = \frac{1}{M}\sum_{s\in I_M} e^{-2ys/M} \le e^{|y|}.$$

Taking products and using $|\ell_{M,\mu}(x+2\pi r+iy)| \le e^{|y_\mu|}$ proves

$$\sum_r |\omega_\mu(x+2\pi r+iy)|^2 \le e^{\sum_\nu |y_\nu|+2|y_\mu|}. \tag{9}$$

The same bound holds for the negative arguments. Cauchy–Schwarz therefore bounds the sum of absolute products in (3) by the same expression. The number of aliases does not appear.

4.2 High aliases and real denominator gaps

Choose $K = \pi + \varepsilon$ with $0 < \varepsilon \le 1/4$. For real $|x_\mu| \le K$, every nonzero alias has a coordinate whose distance from $2\pi M\mathbb{Z}$ is at least $2\pi - K$. Hence

$$P_r(x) \ge H_K := 4(2\pi - K)^2/\pi^2 > 0 \quad (r \ne 0).$$

For complex perturbations,

$$\begin{aligned}\Re P_r(x+iy) &= P_r(x) - 2M^2\sum_\mu \cos((x_\mu+2\pi r_\mu)/M)[\cosh(y_\mu/M)-1]\\ &\ge P_r(x) - e^\delta\sum_\mu y_\mu^2 \qquad (|y_\mu|\le\delta\le 1).\end{aligned}$$

Choose a positive δ so $de^\delta\delta^2 < H_K/2$. All high-alias denominators then have real part at least $H_K/2$. Equation (9) proves that each R_μ is holomorphic and uniformly bounded on this open tube. The central finite averages and P_0 are uniformly bounded there too. For example $|M(e^{iz/M}-1)| \le |z|e^{|\Im z|/M}$.

On the enlarged real cube, set

$$c_K = \frac{\sin(K/2)}{K/2} > 0, \qquad g_* = c_K^{2d+2}, \qquad D_* = 1 + dK^2/H_K.$$

The centered factor $r_M(t) = \sin(t/2)/[M\sin(t/(2M))]$, with $r_M(0) = 1$, satisfies $c_K \le r_M(t) \le 1$ for $|t| \le K$. Using real alias Parseval gives

$$A \ge c_K^{2d}, \qquad g_\mu, D_\mu \ge g_*, \qquad 0 \le R_\mu \le H_K^{-1}, \qquad D_\mu \le D_*. \tag{10}$$

For real nonzero x,

$$E(x) = \frac{\sum_\mu b_\mu(x)/D_\mu(x)}{P_0(x)} \ge \frac{c_K^2}{D_*},$$

since $|d_\mu^+(x)| \ge c_K|x_\mu|$ and $P_0(x) \le |x|^2$. At zero, (7) gives $E(0) = 1$.

4.3 Two successive shrinkings

Take nested real cubes strictly between $[-\pi,\pi]^d$ and $(-K,K)^d$, with nested positive imaginary widths. Cauchy's estimates on the initial tube give uniform derivatives for A, g_μ, D_μ on the first smaller tube. Integrating along $x + ity$ and using (10) keeps these functions uniformly away from zero on a still smaller tube, independently of M.

Formula (7) is now holomorphic and uniformly bounded. Apply Cauchy's estimate once more, using its real lower bound, to keep E uniformly away from zero on a final tube. Its real cube still strictly contains $[-\pi,\pi]^d$. Thus some fixed $\rho_0 > 0$ gives a uniformly bounded holomorphic formula (8) for $|\Im z_\mu| < \rho_0$, with a positive real margin around the central Brillouin cube. The order of these steps avoids assuming the desired nonvanishing of E while estimating it. No Cauchy disk leaves an already controlled open domain.

5 Periodic gluing, kernels, and the reduced inverse

The central-alias split is not individually 2π periodic. The full quotient is periodic on real momenta: shifting by 2π permutes alias labels modulo M, the fine symbols have period $2\pi M$, and the coarse gradient is unchanged. The continuous zero-mode value agrees as well. The analytic extensions on adjacent enlarged Brillouin charts consequently agree on their open real overlaps. Applying the identity theorem successively in the complex coordinates makes them agree holomorphically. The positive real margin is what supplies genuine overlaps. They therefore define one bounded holomorphic periodic symbol on a strip of fixed width.

For its infinite-lattice coefficient

$$q_M(x) = (2\pi)^{-d} \int_{[-\pi,\pi]^d} e^{iz\cdot x} q_M(z)\, dz,$$

shift each contour in the sign of x_μ. For every fixed $0 < \rho < \rho_0$ this gives

$$\|q_M(x)\| \le C_q e^{-\rho|x|_1}.$$

Reducing the exponent gives an absolutely summable weighted kernel. Finite tori evaluate the same variational symbol on their momentum grids, so their kernels are exactly its periodizations. Absolute convergence justifies every image sum. The weighted row sum on a torus is bounded by the infinite weighted row sum because periodic distance is at most the distance of each representative. These bounds include short periods with the correct oriented-cell multiplicities.

5.1 Direct infinite-volume variational identification

The symbol also gives the actual infinite-volume infimum, without interchanging an infimum and a volume limit. For real nonzero momentum let $h = \mathrm{diag}(h_\mu)$ and

$$\theta = \frac{d^- hB}{d^- h d^+}, \qquad B' = B - d^+\theta.$$

The real upper and lower bounds on D_μ imply $|d^+\theta| \le C|B|$ and $|B'| \le C|B|$, uniformly in M. The Feynman minimizing fiber for B' is

$$A_r^F = \mathrm{diag}(\omega_\mu(-p_r))\frac{P_0}{P_r} hB'.$$

For the central alias the ratio is one; for the others it is bounded by $d\pi^2/4$ on the central real cube. Alias Parseval bounds $\sum_r |A_r^F|^2 \le C|B|^2$. Moreover $d^- hB' = 0$ and (2) imply that every A_r^F is fine transverse. Its Maxwell and Feynman energies agree.

Restore the exact coarse datum by adding only the central fine gradient:

$$A_0 = A_0^F + u_0^+\theta/a_M(z), \qquad A_r = A_r^F \ (r \neq 0).$$

Its average is B and its Maxwell energy is unchanged. The added bond field is uniformly bounded because $|u_0^+|/|d^+| \leq \pi/2$ and $|a_M(z)| \geq (2/\pi)^d$. The scalar θ can be singular at zero, but this actual bond field is not. These measurable minimizing fibers define a fine L^2 field for every coarse L^2 field. The zero point has measure zero in infinite volume; on a finite torus use its constant lift. This proves the required variational identification directly.

5.2 The fixed local maps introduce no refinement dependence

T_L uses paths inside one L-block. Each S_L row uses only the two endpoint blocks of its coarse bond. Its center-line interface pivot has discrete coefficient L^{-d}, and no other row uses that pivot. Thus K_L is the identity on retained coordinates and has pivot rows $-L^d(S_L)_{\text{free}}$. It is finite range, with coefficients depending only on fixed d, L.

Group retained coordinates by the L-block containing their bond basepoint. There are $d(L^d-1)$ scalar retained coordinates per block in this convention, with a fixed extra color multiplicity. Since unit-site distance between blocks n, m is at least $L|n-m|_1 - d(L-1)$, the action kernel and these fixed finite-range maps give constants $\eta_0, C_A > 0$, independent of M, with

$$\sum_n e^{\eta_0|n|_1} \, \|A_{M,0}(n)\| \leq C_A. \tag{11}$$

This is the locality input that the spectral gap alone did not supply.

Notebook 06 proves $A_{M,0} \geq \gamma_* I$ uniformly in refinement and volume. Hence the finite-dimensional block Bloch symbol obeys $A_M(\kappa) \geq \gamma_* I$ for every real κ. One can infer this either from the infinite-volume form inequality or from its values on the dense union of finite-torus momentum grids and continuity of the symbol.

By (11), $A_M(\kappa)$ is holomorphic for $|\Im\kappa|_\infty < \eta_0$. For $|y|_\infty \leq \eta_0/2$,

$$\|A_M(x+iy) - A_M(x)\| \leq |y|_\infty \sum_n |n|_1 e^{(\eta_0/2)|n|_1} \, \|A_{M,0}(n)\| \leq C_1|y|_\infty,$$

where C_1 is uniform. Choose a positive $\eta_C < \min(\eta_0/2, \gamma_*/(2C_1))$, omitting the second restriction if $C_1 = 0$. A Neumann inverse about the real symbol gives

$$A_M(\kappa)^{-1} \text{ holomorphic}, \qquad \left\|A_M(\kappa)^{-1}\right\| \leq 2/\gamma_* \quad (|\Im\kappa|_\infty < \eta_C).$$

Contour shifting proves exponential decay of its block kernel and completes Theorem 1.1.

Finite tori sample this same inverse symbol. Their covariance is the periodization of the infinite inverse kernel, rather than the inverse of a truncated infinite covariance. On each fixed region the image error tends to zero exponentially as every period tends to infinity, uniformly in M when comparing the same refinement. No large-period assumption was needed for the uniform bound itself.

6 The free low-momentum normalization

Let $U_\mu(z) = e^{iz_\mu/2}$ and pass to the bond-midpoint convention $q_M^{\text{mid}}(z) = U(-z)q_M(z)U(z)$. The canceled formula becomes

$$(q_M^{\text{mid}})_{\mu\nu}(z) = P_0 h_\mu \delta_{\mu\nu} - \frac{4h_\mu \sin(z_\mu/2)\sin(z_\nu/2)h_\nu}{E}.$$

The functions g_μ, R_μ are even in every momentum coordinate; for R_μ, reflect the corresponding alias index in its symmetric set. The phases cancel in each product of positive and negative arguments. Thus h_μ, E are even too, and $h_\mu(0) = E(0) = 1$, $R_\mu(0) = 0$. Uniform Cauchy bounds in the established neighborhood give

$$h_\mu(z) = 1 + O(|z|^2), \qquad E(z) = 1 + O(|z|^2), \qquad P_0(z) = \sum_\mu z_\mu^2 + O(|z|^4),$$

with constants independent of M. Hence, for small real momentum,

$$q_M^{\mathrm{mid}}(p) = |p|^2 I - pp^T + O(|p|^4).$$

This checks the Maxwell normalization of the flat quadratic model. It is not a statement about an interacting renormalized coupling or the nonlinear continuum limit.

7 The ultraviolet limit of the flat Gaussian baseline

The common complex strip gives more than uniform locality: the flat precision and covariance have limits as the original spacing tends to zero. We prove a quantitative rate, including the aliases near the edge of the fine Brillouin zone. This is a limit of the homogeneous free baseline in the fixed retained coordinates.

For a matrix kernel $F(i,j)$ on a lattice with finite-dimensional fibers, use the exponentially weighted Schur norm

$$\|F\|_\kappa = \max\left\{ \sup_i \sum_j e^{\kappa \operatorname{dist}(i,j)} \|F(i,j)\|, \sup_j \sum_i e^{\kappa \operatorname{dist}(i,j)} \|F(i,j)\| \right\}. \tag{12}$$

The fiber norm is the fixed operator norm. On the unit lattice the distance is $|i-j|_1$. For reduced operators we group all coordinates into the fixed L-blocks and use block distance. Grouping a unit-lattice kernel changes constants and, if needed, reduces its exponent; these changes depend only on d, L and the fixed fiber dimension. Matrix products below use one common grouping. The triangle inequality for distance proves

$$\|FG\|_\kappa \le \|F\|_\kappa \|G\|_\kappa. \tag{13}$$

The same convention applies in periodic distance on a torus, with the actual periodic incidence multiplicities.

Theorem 7.1. *There are a periodic holomorphic symbol q_∞ and constants $\rho_*, C > 0$, independent of odd M, such that*

$$\sup_{|\operatorname{Im} z|_\infty \le \rho_*} \|q_M(z) - q_\infty(z)\| \le C M^{-2}. \tag{14}$$

At a fixed sufficiently small $\kappa > 0$,

$$\|q_M - q_\infty\|_\kappa \le C_\kappa M^{-2}. \tag{15}$$

Set

$$A_{\infty,0} = K_L^*(q_\infty + T_L^* T_L) K_L, \qquad C_{M,0} = A_{M,0}^{-1}, \quad C_{\infty,0} = A_{\infty,0}^{-1}.$$

Then $A_{\infty,0} \ge \gamma_* I$, and at a possibly smaller fixed positive exponent,

$$\|A_{M,0} - A_{\infty,0}\|_\kappa + \|C_{M,0} - C_{\infty,0}\|_\kappa \le C_{d,L,\kappa} M^{-2}. \tag{16}$$

The same bounds hold uniformly on compatible periodic tori. Their kernels are exact periodizations of the infinite kernels. If the shortest nonzero period tends to infinity, the thermodynamic and $M \to \infty$ limits commute on every fixed finite set of coordinates.

7.1 Sinc notation and the exact limiting alias sum

Continue to use $I_M = \{-(M-1)/2, \dots, (M-1)/2\}$ and $p_r(z) = z + 2\pi r$. To distinguish finite and limiting denominators, write the earlier P_r as $P_{M,r}$. Define

$$s_M(w) = \frac{\sin(w/2)}{M \sin(w/(2M))}, \qquad s(w) = \frac{\sin(w/2)}{w/2}, \qquad s(0) = 1, \tag{17}$$

with all removable values filled, and

$$P_{M,r} = 4M^2 \sum_\nu \sin^2 \frac{p_{r,\nu}}{2M}, \qquad P_{\infty,r} = \sum_\nu p_{r,\nu}^2. \tag{18}$$

The paired averaging numerators are

$$N_{M,r,\mu} = s_M(p_{r,\mu})^2 \prod_\nu s_M(p_{r,\nu})^2, \tag{19}$$

$$N_{\infty,r,\mu} = s(p_{r,\mu})^2 \prod_\nu s(p_{r,\nu})^2. \tag{20}$$

Indeed the longitudinal averaging phase $\exp(i(M-1)p_\mu/(2M))$ tends to $\exp(ip_\mu/2)$, but it cancels exactly between the positive and negative arguments of each covariance numerator. There is consequently no phase error of order M^{-1}. These are analytic products, not complex absolute squares. Physical-volume Parseval normalization gives no additional power of M in the alias sum.

Define the high-alias remainders

$$R_{M,\mu} = \sum_{r \in I_M^d \setminus \{0\}} \frac{N_{M,r,\mu}}{P_{M,r}}, \qquad R_{\infty,\mu} = \sum_{r \in \mathbb{Z}^d \setminus \{0\}} \frac{N_{\infty,r,\mu}}{P_{\infty,r}}. \tag{21}$$

We first prove normal convergence and a uniform rate for the second sum. These estimates also justify the explicit continuum formula below.

7.2 High-alias bounds uniform up to the fine cutoff

Fix $\pi < K < 2\pi$ and work on

$$\mathcal{B}_{K,\rho} = \{x + iy : |x_\nu| \le K,\ |y_\nu| \le \rho\}, \qquad 0 < \rho \le 1.$$

Put $a = 2\pi - K$ and $c_0 = 4a^2/\pi^2$. If a retained component $r_\nu \ge 1$, the two possible nearest distances of $x_\nu + 2\pi r_\nu$ to $2\pi M\mathbb{Z}$ obey

$$2\pi r_\nu - K \ge a r_\nu, \qquad 2\pi(M - r_\nu) - K \ge 2\pi(r_\nu + 1) - K \ge a r_\nu.$$

The negative case is symmetric. The principal distance is at most πM, so the elementary sine lower bound gives

$$P_{M,r}(x) \ge c_0 |r|_2^2 \qquad (r \ne 0).$$

Moreover, direct expansion of the complex sine gives

$$\operatorname{Re} P_{M,r}(x + iy) \ge P_{M,r}(x) - e^\rho \sum_\nu y_\nu^2.$$

Choose ρ so that $de^\rho \rho^2 \le c_0/2$. Then

$$|P_{M,r}(z)| \ge \operatorname{Re} P_{M,r}(z) \ge \frac{c_0}{2} |r|_2^2 \qquad (r \ne 0). \tag{22}$$

This bound includes $|r_\nu|$ comparable to M. The continuum denominator, for every nonzero integer alias, satisfies

$$\operatorname{Re} P_{\infty,r}(z) = \sum_\nu (x_\nu + 2\pi r_\nu)^2 - \sum_\nu y_\nu^2 \geq \frac{c_0}{2}|r|_2^2. \tag{23}$$

For this last inequality use $|x_\nu + 2\pi r_\nu| \geq a|r_\nu|$ when $r_\nu \neq 0$, and $a^2 \geq c_0$.

The integral representation of sinc bounds the zero-index factor by $e^{\rho/2}$. For $r_\nu \neq 0$,

$$|\sin((z_\nu + 2\pi r_\nu)/2)| \leq \cosh(\rho/2), \qquad |z_\nu + 2\pi r_\nu| \geq a|r_\nu|.$$

Hence

$$|s(p_{r,\nu})| \leq \frac{C_s}{1 + |r_\nu|}. \tag{24}$$

The same estimate holds for retained s_M factors. For a nonzero index, use the preceding distance estimate to obtain

$$|M \sin(p_{r,\nu}(z)/(2M))| \geq |M \sin((x_\nu + 2\pi r_\nu)/(2M))| \geq (a/\pi)|r_\nu|;$$

for zero index, use the centered finite exponential average.

Set

$$\mathcal{W}_{r,\mu} = (1 + |r_\mu|)^{-2} \prod_\nu (1 + |r_\nu|)^{-2}, \qquad B_2 = \sum_{m \in \mathbb{Z}} (1 + |m|)^{-2} = \pi^2/3 - 1.$$

Equations (22)–(24) give the summable majorant

$$\left|\frac{N_{M,r,\mu}}{P_{M,r}}\right| + \left|\frac{N_{\infty,r,\mu}}{P_{\infty,r}}\right| \leq C\frac{\mathcal{W}_{r,\mu}}{|r|_2^2}. \tag{25}$$

The finite term here is used only for retained aliases. In particular, the infinite remainder is holomorphic in the box interior and continuous uniformly on the closed box. Its tail has the explicit bound

$$\sum_{|r|_\infty > N} \frac{\mathcal{W}_{r,\mu}}{|r|_2^2} \leq \frac{2dB_2^{d-1}}{(N+1)^3}. \tag{26}$$

To verify it, discard the additional orientation factor, use $|r|_2 \geq N + 1$, and cover the tail by the d coordinate tails; $\sum_{|m|>N}(1 + |m|)^{-2} \leq 2/(N + 1)$.

7.3 A second-order rate without a small-alias restriction

Let $f(w) = w/\sin w$, with its removable value at zero. For all retained aliases and $M \geq 3$,

$$\left|\operatorname{Re}\frac{p_{r,\nu}}{2M}\right| \leq \frac{\pi}{2} + \frac{K - \pi}{6} < \frac{2\pi}{3}, \qquad \left|\operatorname{Im}\frac{p_{r,\nu}}{2M}\right| \leq \frac{1}{6}. \tag{27}$$

Thus f and $(f(w) - 1)/w^2$ are bounded on one fixed rectangle, after filling the removable value at zero. The exact identity

$$s_M(p) = s(p)f(p/(2M)), \qquad |f(p/(2M)) - 1| \leq C|p|^2/M^2 \tag{28}$$

holds even at the largest aliases. For $r \neq 0$, $\sum_\nu |p_{r,\nu}|^2 \leq C|r|_2^2$. Telescoping the fixed $2d + 2$ factors in the numerator therefore gives

$$|N_{M,r,\mu} - N_{\infty,r,\mu}| \leq CM^{-2}|r|_2^2\mathcal{W}_{r,\mu}. \tag{29}$$

The removable analytic function $((\sin w/w)^2-1)/w^2$ is bounded on the same rectangle, yielding

$$|P_{M,r}-P_{\infty,r}|\le CM^{-2}\sum_{\nu}|p_{r,\nu}|^4\le CM^{-2}|r|_2^4. \tag{30}$$

Divide (29) by the lower denominator bound. For the reciprocal-denominator difference, divide (30) by the product of the two lower bounds. The powers of $|r|$ cancel in both operations, giving

$$\left|\frac{N_{M,r,\mu}}{P_{M,r}}-\frac{N_{\infty,r,\mu}}{P_{\infty,r}}\right|\le CM^{-2}\mathcal{W}_{r,\mu}. \tag{31}$$

Since $\sum_r \mathcal{W}_{r,\mu}\le B_2^d$, these retained-alias errors sum to $O(M^{-2})$. The missing aliases are bounded by (26) with $N=(M-1)/2$, giving $O(M^{-3})$. Consequently

$$\sup_{z\in\mathcal{B}_{K,\rho}}|R_{M,\mu}(z)-R_{\infty,\mu}(z)|\le CM^{-2}. \tag{32}$$

No estimate above assumes $|r|/M$ is small. The case $M=1$ can be included by enlarging the constant; it has no high aliases.

7.4 The canceled continuum symbol and its removable zeros

To avoid confusion with the reduced precision $A_{M,0}$, denote the principal scalar averaging weight in this subsection by $\mathfrak{a}_M$. Define its limit and the other limiting quantities by

$$P(z)=\sum_{\nu}z_\nu^2,\qquad \mathfrak{a}(z)=\prod_{\nu}s(z_\nu)^2,\qquad g_{\infty,\mu}(z)=\mathfrak{a}(z)s(z_\mu)^2, \tag{33}$$
$$D_{\infty,\mu}=g_{\infty,\mu}+PR_{\infty,\mu},\qquad h_{\infty,\mu}=D_{\infty,\mu}^{-1}.$$

Here $\mathfrak{a}_M(z)=\prod_\nu s_M(z_\nu)^2$ is the quantity previously called the central $A(z)$, not a retained-coordinate matrix. Keep $d_\mu^{\pm}=e^{\pm iz_\mu}-1$ and $b_\mu=d_\mu^+d_\mu^-$, and set

$$E_\infty=\mathfrak{a}^{-1}-\sum_{\mu}\frac{b_\mu R_{\infty,\mu}}{g_{\infty,\mu}D_{\infty,\mu}}. \tag{34}$$

The explicit limiting precision is

$$(q_\infty)_{\mu\nu}(z)=P(z)h_{\infty,\mu}(z)\delta_{\mu\nu}-\frac{h_{\infty,\mu}(z)d_\mu^+(z)d_\nu^-(z)h_{\infty,\nu}(z)}{E_\infty(z)}. \tag{35}$$

On real nonzero momenta, this is the gauge quotient of the continuum alias covariance with diagonal entries

$$c_{\infty,\mu}(x)=\sum_{r\in\mathbb{Z}^d}\frac{s(x_\mu+2\pi r_\mu)^2\prod_\nu s(x_\nu+2\pi r_\nu)^2}{\sum_\nu(x_\nu+2\pi r_\nu)^2}. \tag{36}$$

Explicitly, if $W_\infty=\mathrm{diag}(c_{\infty,\mu})$, it equals $W_\infty^{-1}-W_\infty^{-1}d^+(d^-W_\infty^{-1}d^+)^{-1}d^-W_\infty^{-1}$. The singular principal Feynman term of (36) is not used at $x=0$; the canceled formula (35) supplies the continuation.

Choose a closed smaller box inside the established common holomorphic domain, still with a real margin beyond $[-\pi,\pi]^d$. The finite denominators $g_{M,\mu}, D_{M,\mu}, E_M$ have uniform positive modulus lower bounds there. At the central alias, (28) and the direct sine expansion give $O(M^{-2})$ convergence of $P_{M,0}$ to P and $\mathfrak{a}_M, g_{M,\mu}$ to their limits. Together with (32), this gives $O(M^{-2})$ convergence of $D_{M,\mu}$ and then E_M. Their lower bounds pass to the limits. The canceled rational expression is uniformly Lipschitz on this bounded set separated from its denominator zeros. This proves (14) on the smaller strip.

Every one-variable zero at $z_\nu + 2\pi r_\nu = 0$ is filled by the entire sinc function. If $z_\nu = 0$ and $r_\nu \neq 0$, that sinc factor is zero and the alias term vanishes; its high denominator is still nonzero by (23). At $z = 0$, all high numerators vanish, while

$$\mathfrak{a}(0) = g_{\infty,\mu}(0) = D_{\infty,\mu}(0) = E_\infty(0) = 1, \qquad P(0) = d^{\pm}(0) = 0.$$

Thus $q_\infty(0) = 0$. The exact identity $\sum_\mu b_\mu / g_{\infty,\mu} = P/\mathfrak{a}$ holds with the removable one-variable factors filled. It removes the whole complex null cone $P = 0$ from (35). There is no exceptional rule on coordinate axes.

The separated central and remainder pieces need not be periodic. The full finite-M quotients are periodic on their common tube, so their uniform limit is periodic. Alternatively, reindex the absolutely convergent full sum (36) on the real overlaps and apply the identity theorem. This also identifies the formula across Brillouin-chart seams.

7.5 Weighted convergence, inversion, and compatible tori

Choose $0 < \kappa < \rho_0 < \rho_*$. In the Fourier convention used above, shift the νth contour by $i\rho_0 \operatorname{sign}(x_\nu)$. Equation (14) gives

$$\|(q_M - q_\infty)(x)\| \le C M^{-2} e^{-\rho_0 |x|_1}.$$

Summation proves (15), with the explicit factor

$$\|q_M - q_\infty\|_\kappa \le \frac{C}{M^2} \left(\frac{1 + e^{-(\rho_0 - \kappa)}}{1 - e^{-(\rho_0 - \kappa)}} \right)^d. \tag{37}$$

Grouping into the fixed L-block fibers preserves this rate. Because K_L, T_L are fixed finite-range maps,

$$A_{M,0} - A_{\infty,0} = K_L^*(q_M - q_\infty)K_L, \qquad \|A_{M,0} - A_{\infty,0}\|_\kappa \le \|K_L\|_\kappa^2 \, \|q_M - q_\infty\|_\kappa .$$

This is an operator-norm limit as well. Hence the real lower gap $A_{M,0} \ge \gamma_* I$ passes to $A_{\infty,0}$. The uniform locality and gap argument already proved for the flat covariance applies to the limit too. Reducing the exponential weight below the pointwise inverse-kernel decay exponent yields

$$\sup_M \|C_{M,0}\|_\kappa + \|C_{\infty,0}\|_\kappa < \infty.$$

The exact resolvent identity and (13) then give

$$C_{M,0} - C_{\infty,0} = C_{M,0}(A_{\infty,0} - A_{M,0})C_{\infty,0},$$

$$\|C_{M,0} - C_{\infty,0}\|_\kappa \le \|C_{M,0}\|_\kappa \, \|A_{M,0} - A_{\infty,0}\|_\kappa \, \|C_{\infty,0}\|_\kappa \le C_{d,L,\kappa} M^{-2}.$$

This completes the infinite-volume estimate. In the sequence $M = L^k$ its rate is $O(L^{-2k})$.

Let Γ be a compatible full-rank period lattice. Unit-lattice symbols sampled on the torus Fourier grid have periodized kernels, so

$$(q_M^\Gamma - q_\infty^\Gamma)(x, y) = \sum_{\gamma \in \Gamma} (q_M - q_\infty)(x - y + \gamma). \tag{38}$$

The periodic distance is no greater than the distance of each image; thus its weighted Schur norm is bounded by (37), independently of volume.

For retained operators require $\Gamma \subset L\mathbb{Z}^d$ and the same periodic pivot convention. In L-block coordinates all operators have fixed finite-dimensional fibers. Periodization commutes with multiplication of absolutely summable block kernels: expand both image sums and decompose the intermediate point into

a fundamental-domain point and its period translate. Therefore periodizing $A_{M,0}C_{M,0} = I = C_{M,0}A_{M,0}$ proves

$$C^{\Gamma}_{M,0} = (A^{\Gamma}_{M,0})^{-1},$$

with $A^{\Gamma}_{M,0}$ exactly the finite periodic source matrix. The same reasoning holds for the limit. This verifies the volume-uniform version of (16), including short tori with coincident images and their correct multiplicities.

If the shortest nonzero period tends to infinity, uniform exponential decay makes the image error vanish on every fixed finite coordinate set. Combining this error with the volume-independent M^{-2} bound proves that the two limits commute. Growing volume with a bounded periodic direction instead produces a cylindrical limit. The proof uses exact homogeneous periodization; it says nothing about an arbitrary Dirichlet or retained-history covariance being a compression or periodization of the homogeneous one.

Remark 7.2. The limit is the order-zero homogeneous Gaussian baseline. Under the finite-level constrained-minimizer identification, it is expressed in the actual retained integration coordinates of the source step. It does not establish an interacting Yang–Mills ultraviolet limit, nonflat-background convergence, independent-current derivative bounds, or arbitrary retained-history boundary matching. Those interfaces are unchanged, and no sharp cutoff domain is moved by the calculation.

8 The actual flat source covariance

The maps must remain distinct. Q^{st}_M goes from original spacing M^{-1} to unit spacing; the rooted composition has the same energy infimum through the explicit gradient adjustment of Notebook 06. S_L, T_L, K_L concern the next unit-to-L step. The averaged Feynman covariance W is an intermediate calculation, not the actual reduced fluctuation covariance.

Raw II, the small-field step around lines 73860–74020, records the quadratic expansion, exponential gauge fixing, straightening of the nonlinear constraint, coupling rescaling, and pivot elimination of CMP109 (2.11)–(2.13). At $(U, J) = (1, 0)$, the stationary action Hessian is q_M, the gauge Hessian is $T^*_L T_L$, and the constraint straightening starts at quadratic order. Its second derivative therefore does not change the stationary Hessian. Haar and Jacobian terms remain in the fluctuation interaction. The retained Gaussian coordinates are exactly the nonpivot variables inserted by K_L. Using the constrained-minimizer identification of Notebook 06, whose homogeneous smoothness premise is now supplied by Notebook 15 and matched explicitly in Notebook 17, the source covariance is

$$C^{(k)}(1) = \big(K^*_L[q_M + T^*_L T_L]K_L\big)^{-1}.$$

The kernel convention matters: Raw I, lines 46960–47000, writes the discrete next-step matrix coefficient as $L^{-d}Q^{\flat}(c, b)$. At flatness its pivot kernel is the identity, so the actual insertion coefficient is $-Q^{\flat}(c, b) = -L^d S_L(c, b)$ on nonpivot columns, exactly as used here. No nonlocal inverse is hidden in this pivot map.

The source's differently gauge-fixed auxiliary G_1 is not the Feynman inverse used in (3). Its transcribed identities $G_1^{-1} = \Delta_1 + DRD^* + Q^* aQ$, $H_1 = G_1 Q^*(QG_1Q^*)^{-1}$, $QH_1 = I$, and $RD^*H_1 = 0$ instead give $H_1^*\Delta_1 H_1 = (QG_1Q^*)^{-1} - a$. The variational source match identifies that physical action Hessian with q_M; it does not identify the two auxiliary propagators.

The result supplies a uniformly positive, bounded, exponentially local flat covariance on the exact retained coordinates of this full-block step. What remains for the Gaussian cutoff application is weighted control of the actual complex independent-current precision perturbation or its source responses. The flat proof does not provide those derivatives, arbitrary-reference background locality, mixed-history domain bounds, or permission to change a sharp cutoff's integration coordinates. Overlapping interactions and the continuum Yang–Mills obligations also remain separate.

Notebook 08 constructs a fine minimizing extension with uniform exponential locality in the deposited homogeneous flat source gauge. It controls the potential and first fine derivatives in physical Hölder norms below exponent one, the curl in the corresponding Hölder norm, and the componentwise Laplacian in supremum norm. This supplies local flat external legs for subsequent coefficient variations. It does not by itself bound the actual nonflat or independent-current perturbation of the precision.

References

[1] Pinned Yang–Mills-LLM-experiment account, commit `e392ef45013a581664bab76b90867170b30fe10e`.

[2] T. Bałaban, *Renormalization group approach to lattice gauge field theories I*, Commun. Math. Phys. 109 (1987), 249–301. Copy of the original article.